NAME: ALLAN NJUGUNA NGUGI
EMAIL: ngugiallan05@gmail.com
ADMISSION NUMBER: CS-SA12-26007

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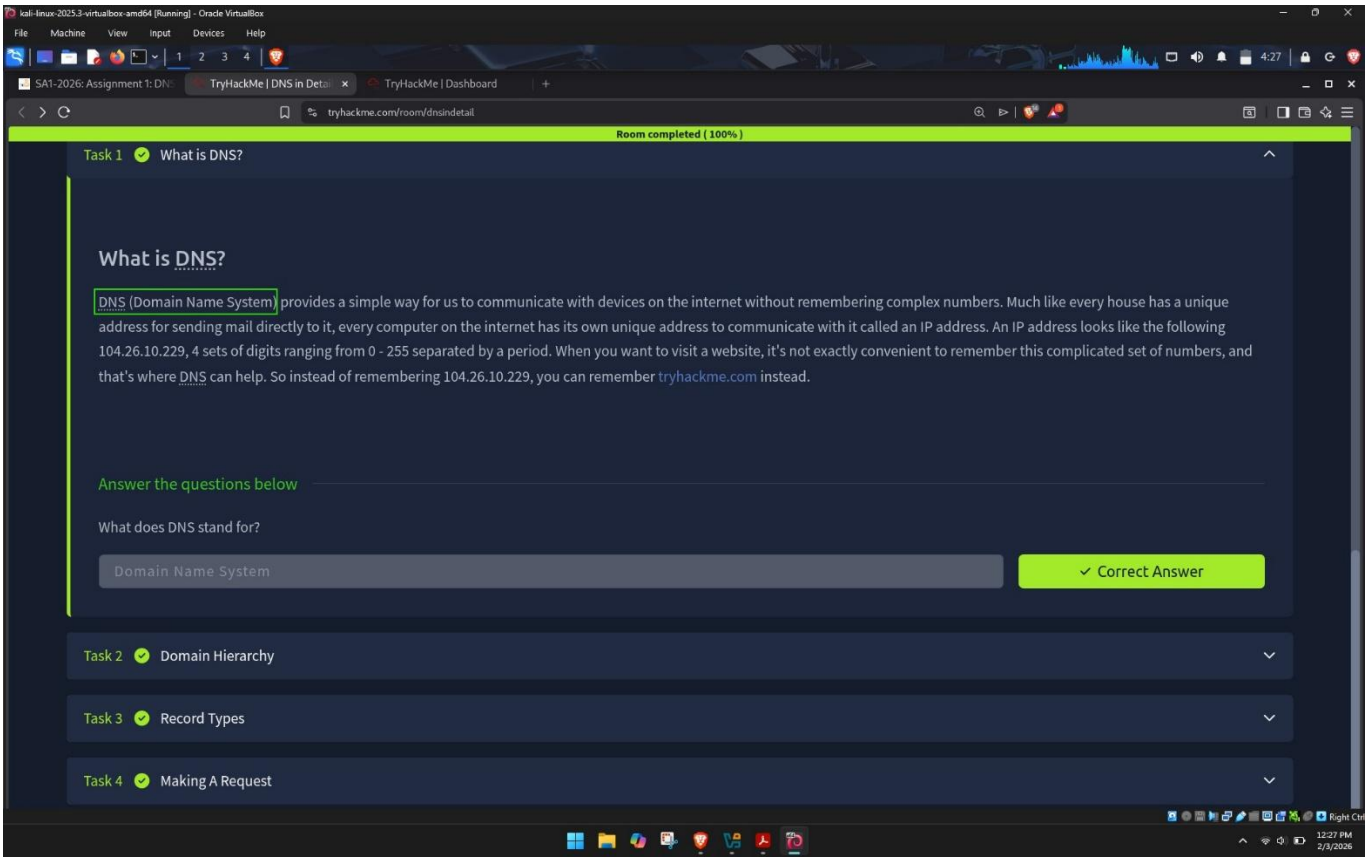
Introduction

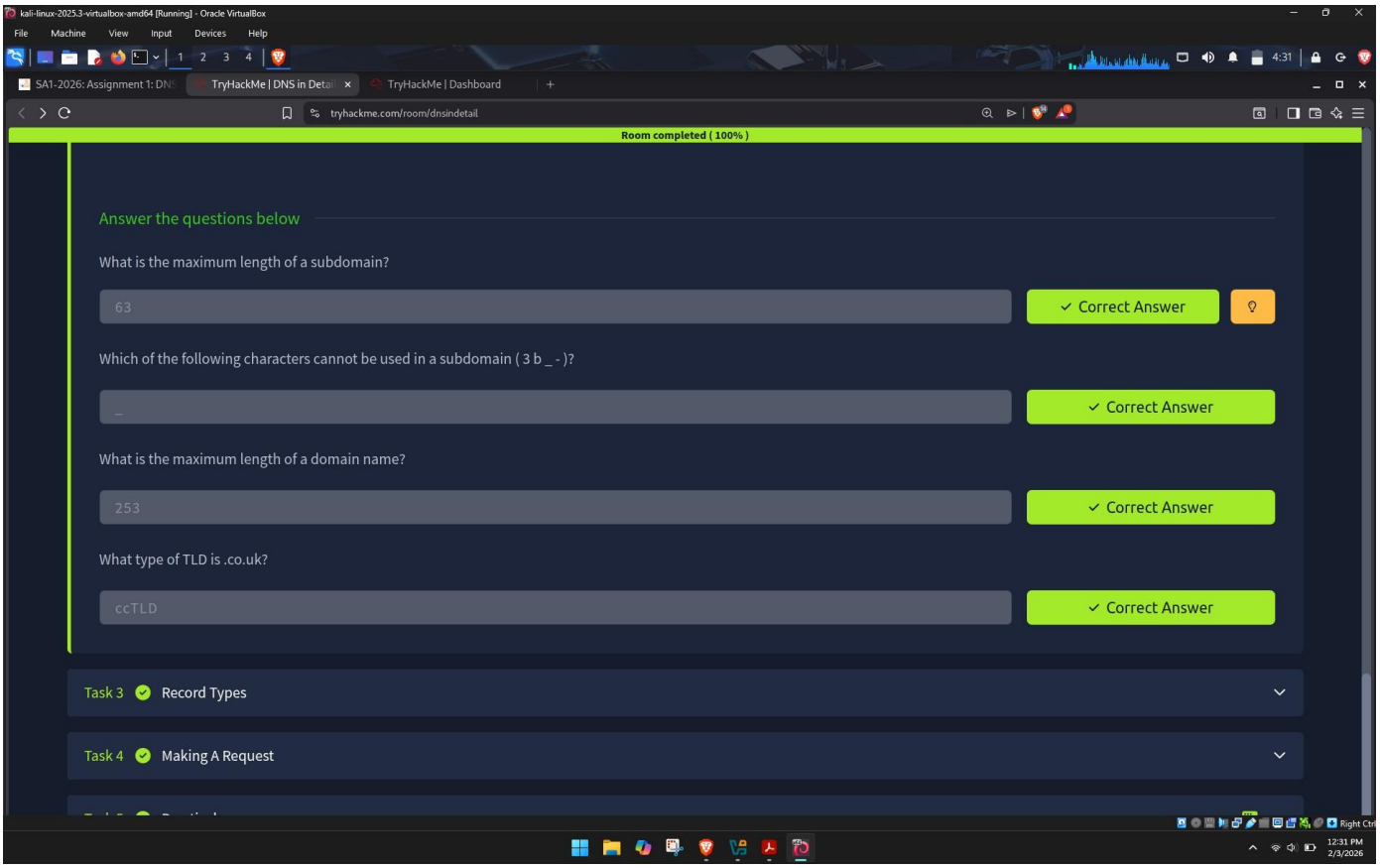
This room covered the essentials of DNS (Domain Name System) which is a protocol that operates as a digital address book because it converts user-friendly domain names such as tryhackme.com into the sophisticated IP addresses that computers use for identification purposes. The internet assigns each device a distinct IP address that consists of four numerical groups which are separated by period marks, while DNS makes browsing easier because it eliminates the need for users to remember these numerical sequences.



Task 2: Domain Hierarchy

I learned that Top-Level Domains (TLDs) are the suffixes like .com or .gov that categorize a site by purpose or location. I also learned that the Second-Level Domain is the main identifiable name (like "tryhackme"), while subdomains are prefixes added to the left to further organize the

site. The study revealed that both second-level domains and subdomains must adhere to particular character restrictions while their total length cannot exceed 253 characters.

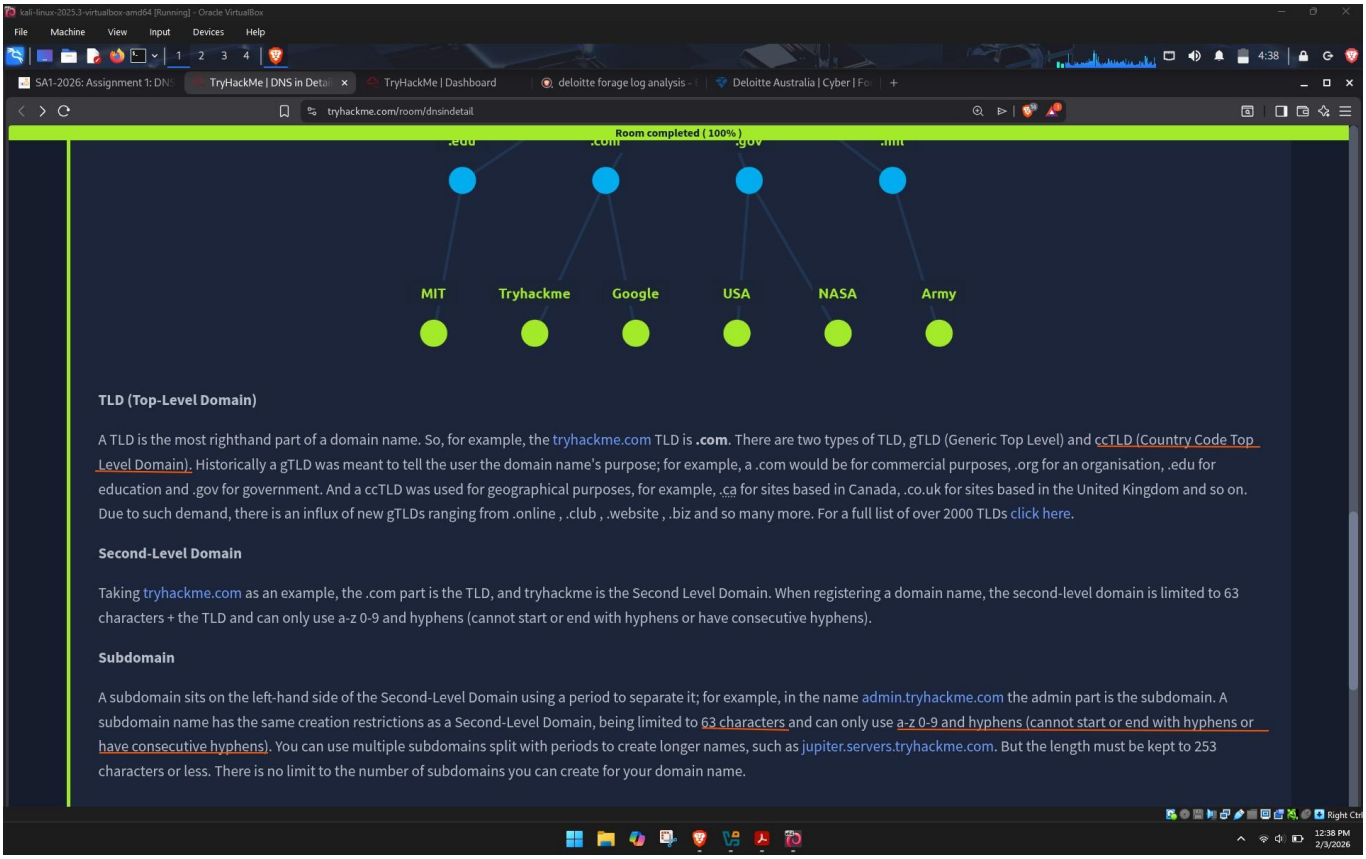


The first question inquired on the maximum length of a subdomain, which the answer was 63, and this was derived in the text below the title subdomain, as shown in the screenshot below:

The second question inquired on the character that cannot be used in a subdomain: and options were given. The correct option was the underscore character, since the only characters that are utilized are a to z, 0 to 9 and hyphens. This information is derived in the screenshot below.

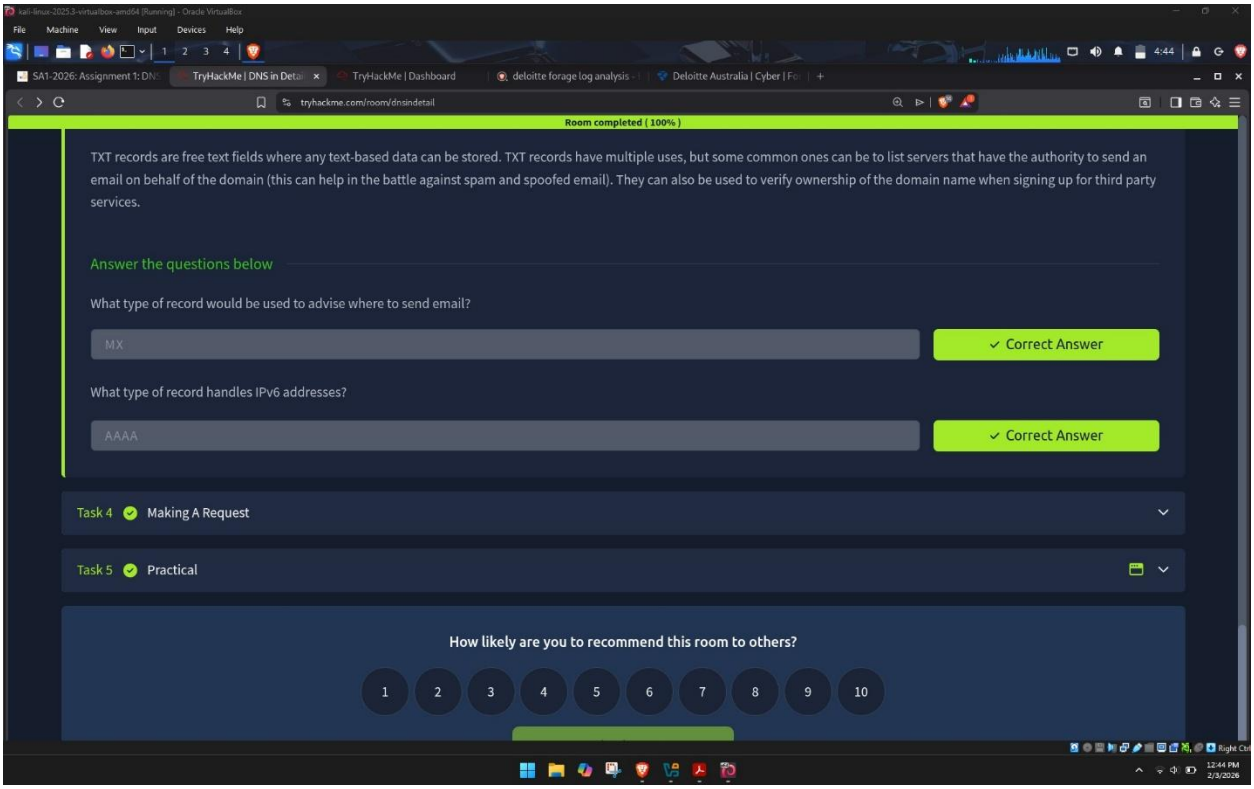
The third question inquired on the maximum length of a domain name, which is 253 characters. The length has to be kept to 253 characters, as shown in the image below.

The last question in this task focuses on the type of top-level domain tied to ‘.co.uk’. This is a top- level domain that is tied to a country code (The UK) which then qualifies it as a country code TLD. This is shown in the image below.



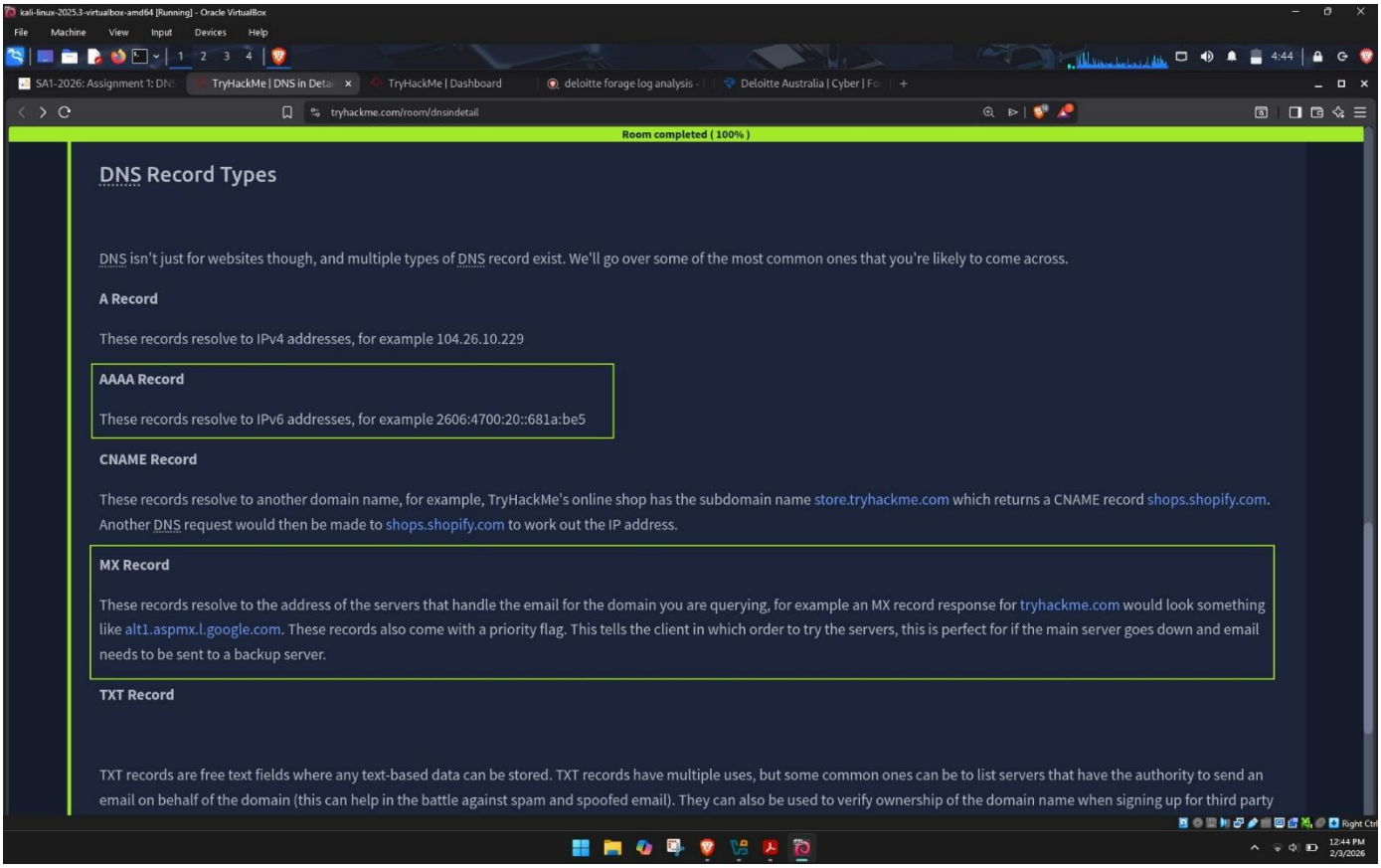
Task 3: Record Types

This task touched on the various existent DNS records; which are the A record, the AAAA record, the CNAME record, the MX record and the TXT record.



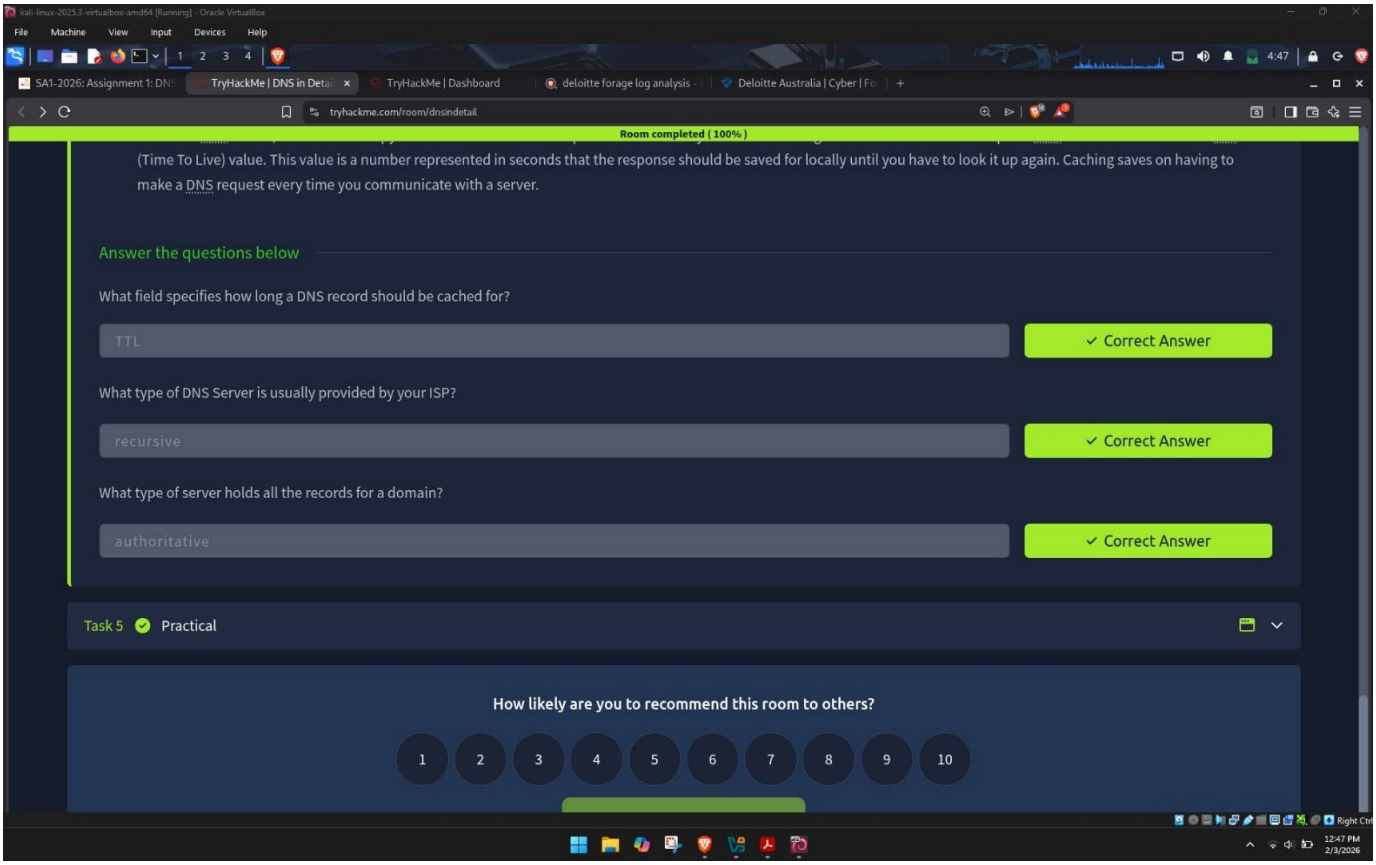
The first question touches on the record that advices where to send email; which is the MX record. This is shown in the image below.

The second question also touches on another DNS record, but one that handles IPV6 addresses; which is the AAAA record. This is shown in the image below:



Task 4: Making a request

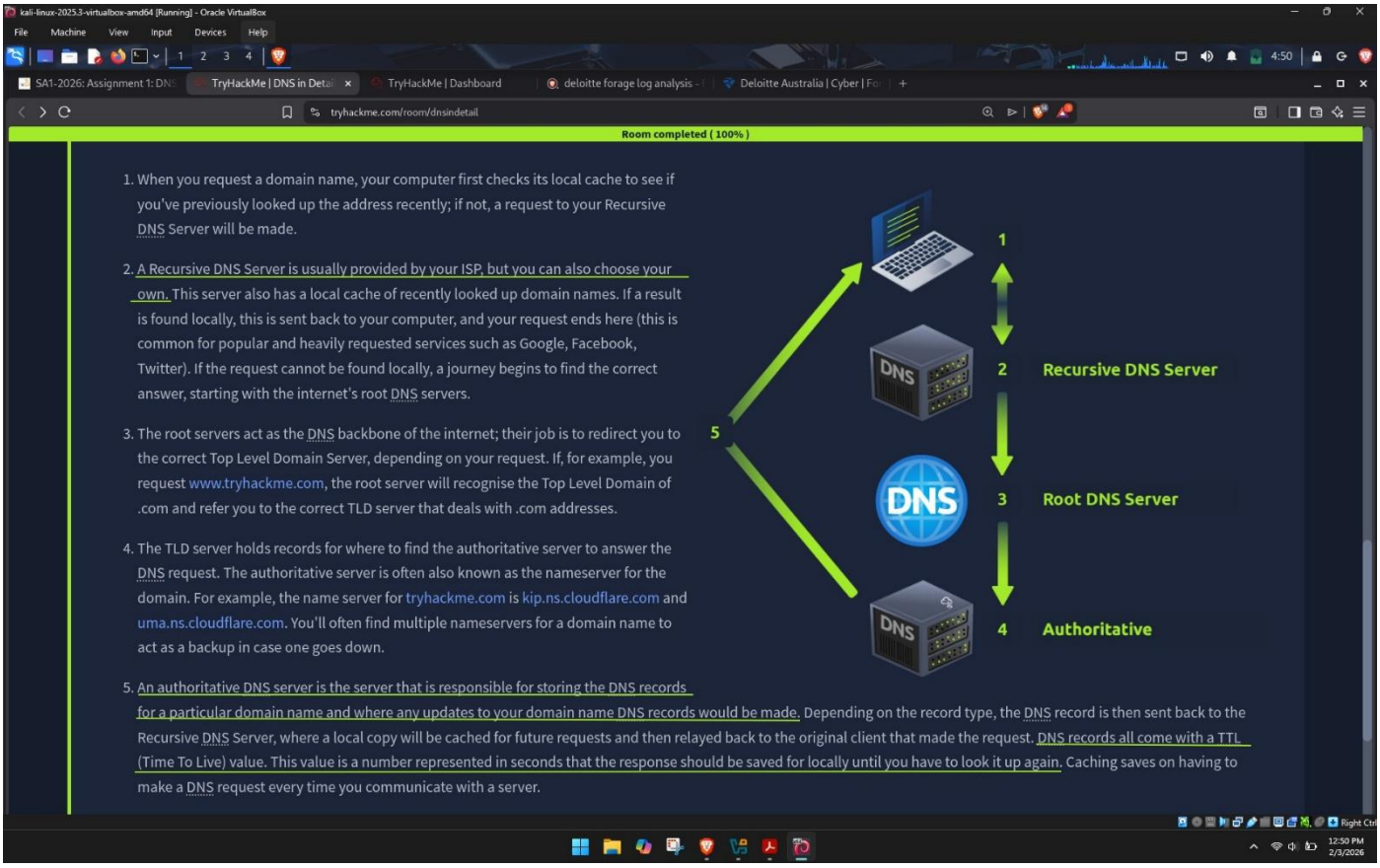
I learned that when a domain request isn't found in a local cache, a Recursive DNS Server queries the internet's hierarchy—starting from Root servers to TLD servers—to locate the Authoritative DNS server holding the specific records. The system sends the discovered IP address back to the user while storing it in a cache which will last until the Time To Live (TTL) value expires to enhance performance during subsequent visits.



The first question touched on the field that specifies how long a DNS record should be cached for, which is TTL (Time To Live).

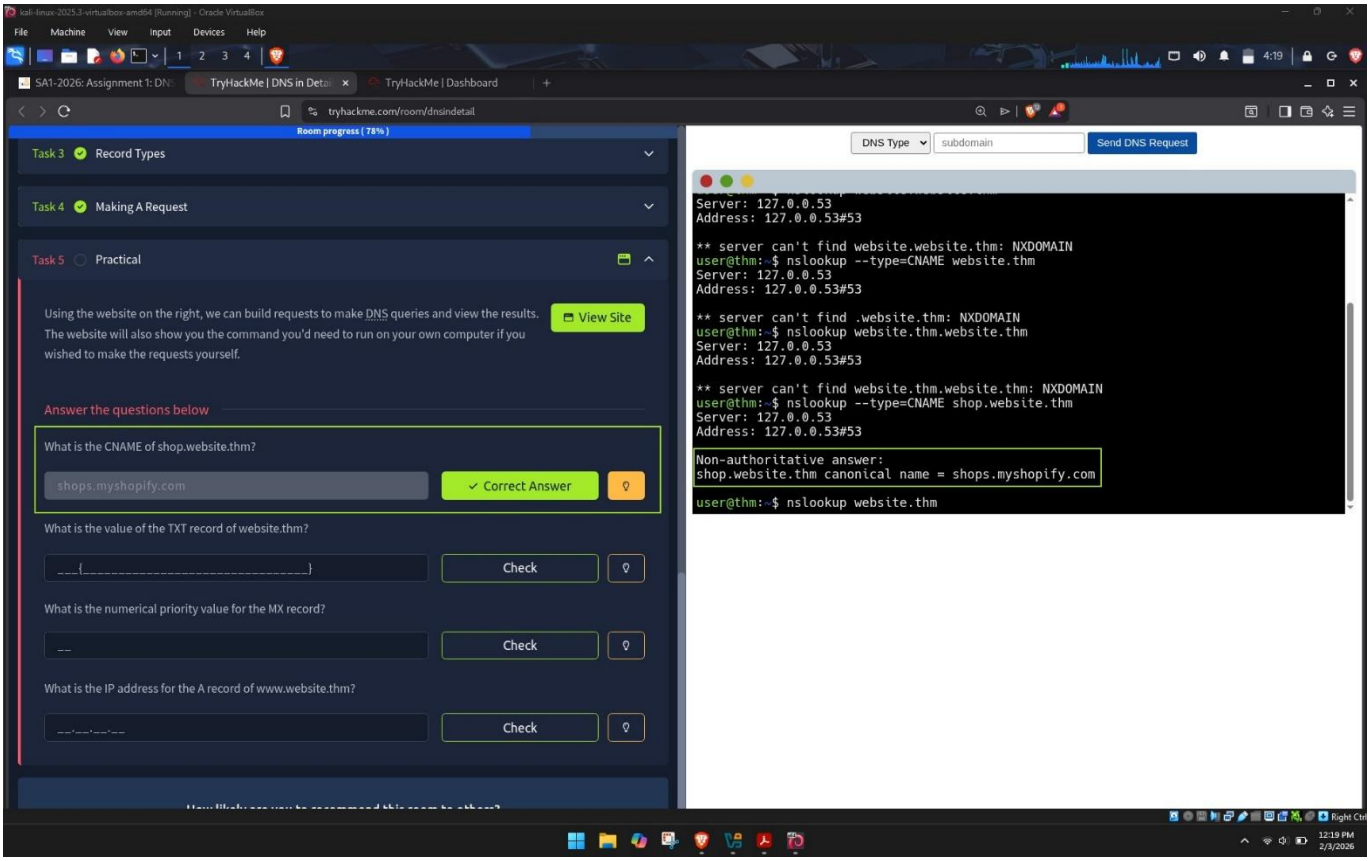
The second question focuses on the type of DNS server which is usually provided by one's ISP, which is recursive.

The last question touches on the type of server that holds all the records for a domain, which is authoritative; as shown in the image below:

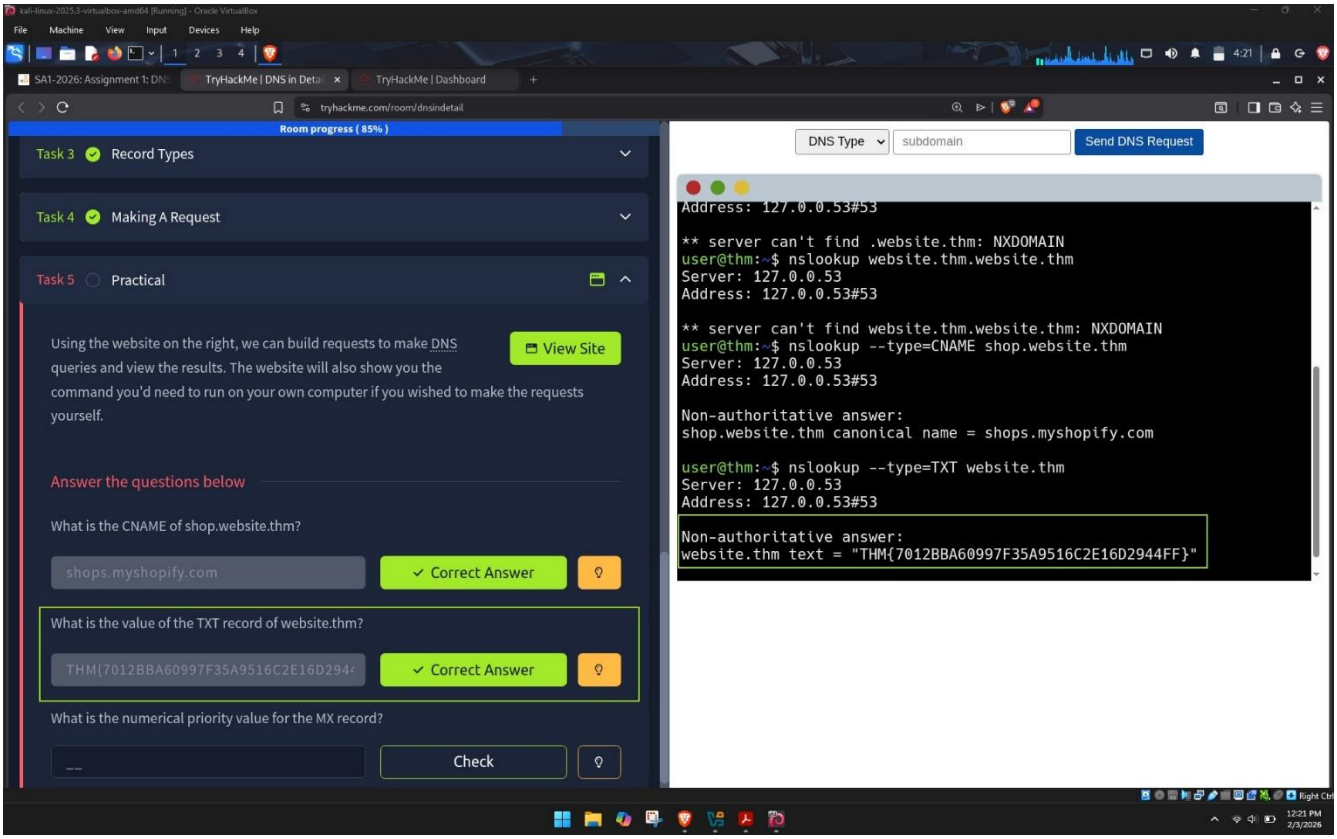


Task 5: Practical

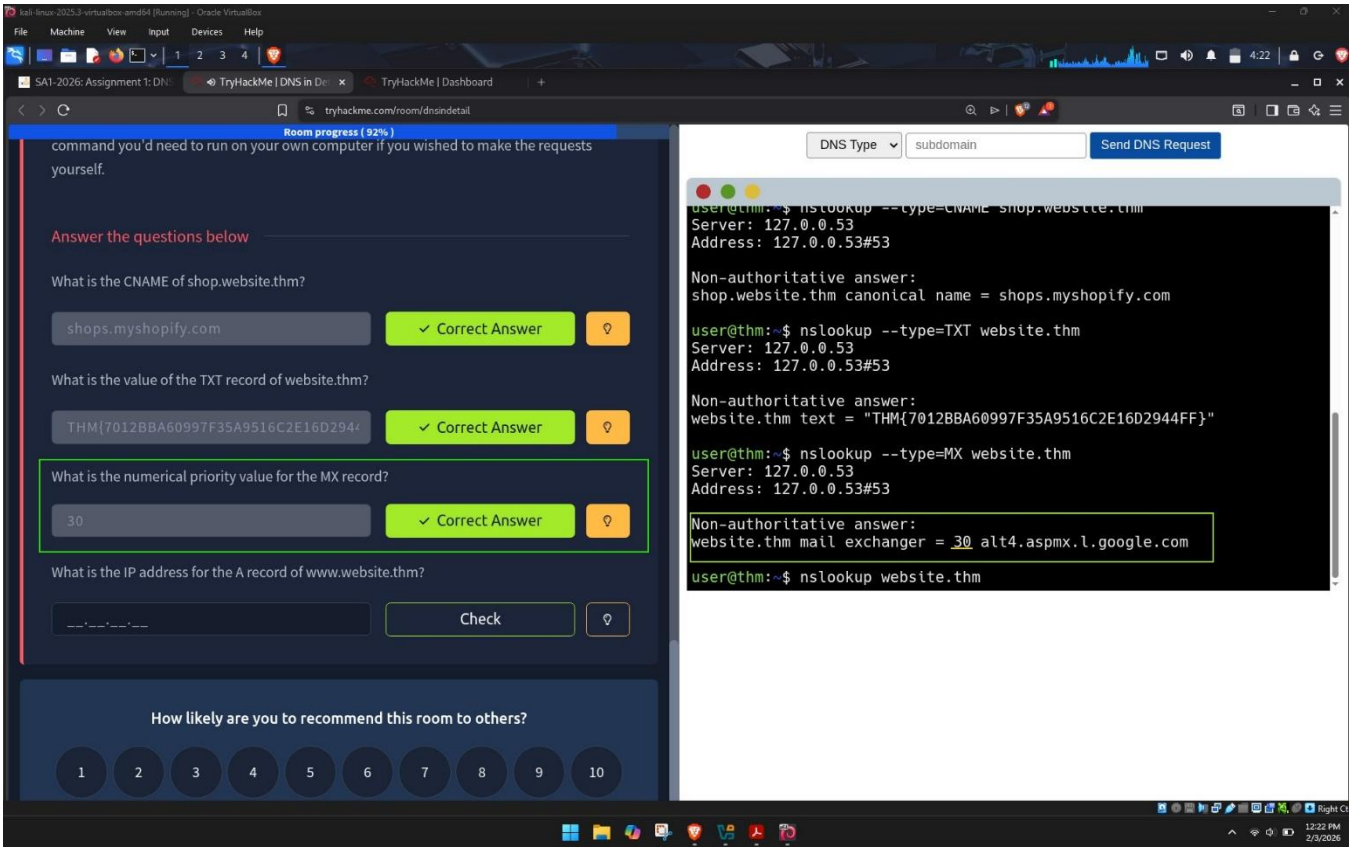
The task focused on a certain website, which required one to build requests to make DNS queries to view the results.



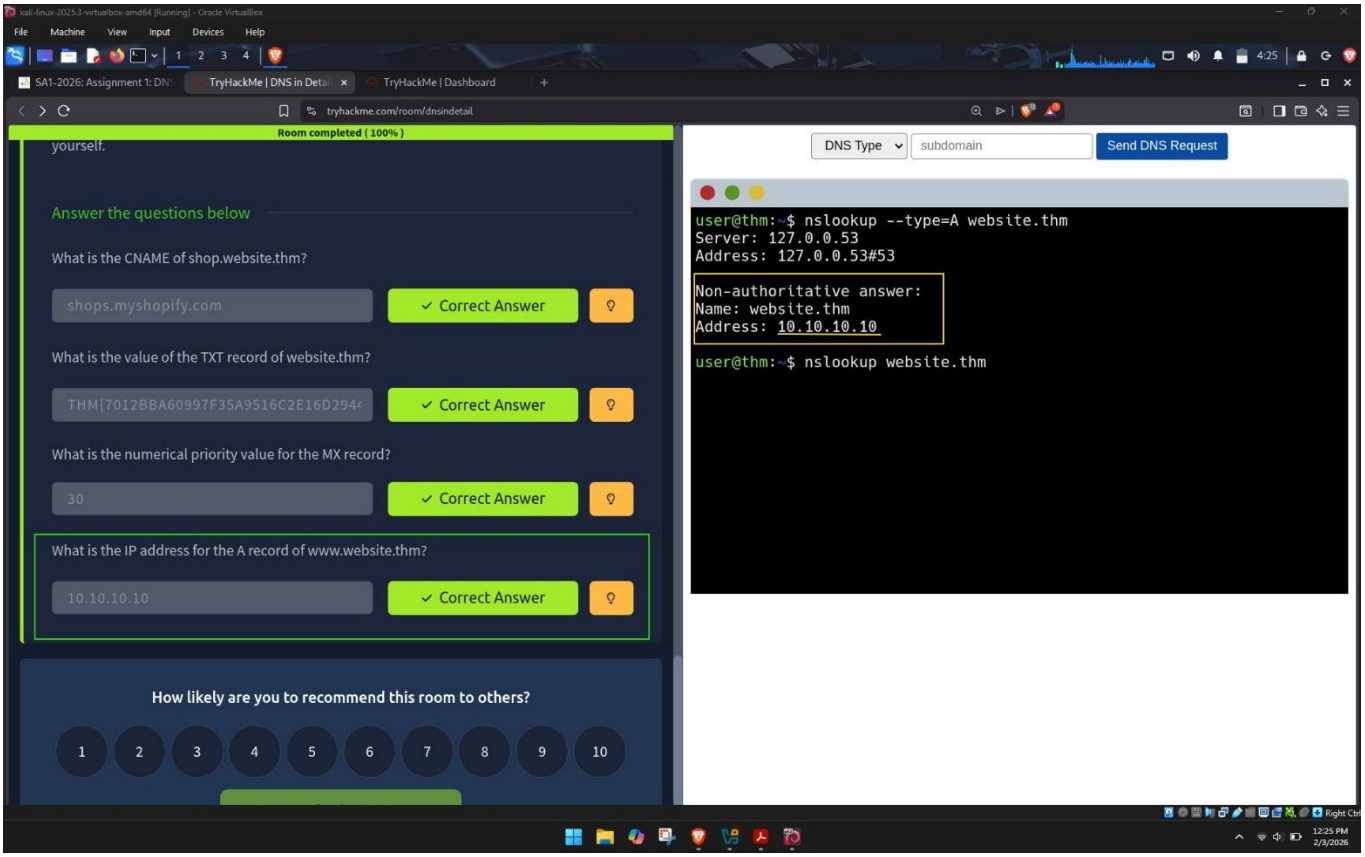
In order to get the answer to the first question, one had to set the DNS type to CNAME and set the subdomain name to shop. The output is shown above.



In order to get the answer, one had to set the DNS type to TXT and subdomain to website, which the output is a flag as the answer.

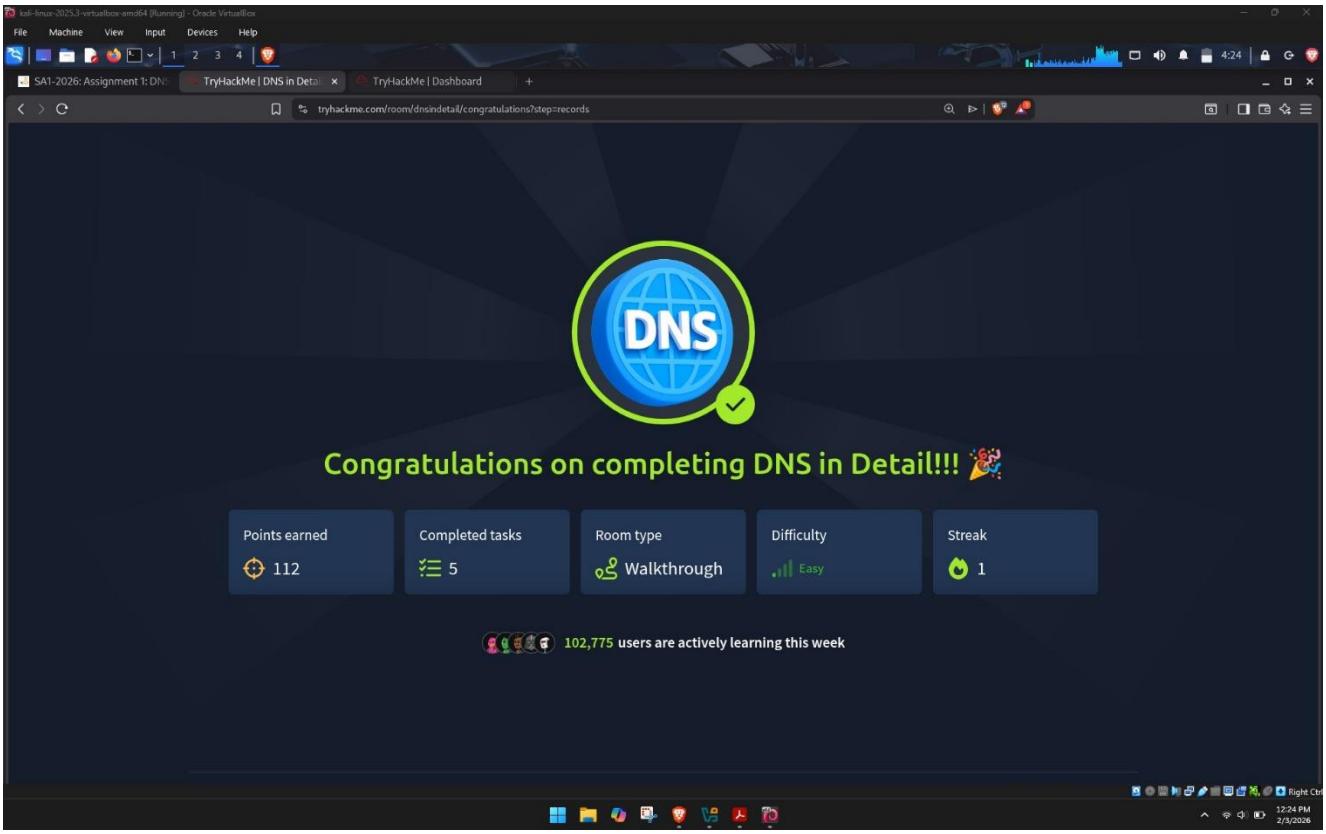


The output is derived by setting the DNS type to MX, and the output is shown in the image above.



The output is derived by setting the DNS type to A, and the output is shown above.

Screenshot Showing Module Completion



Conclusion

The Domain Name System functions as the fundamental directory system for the internet because it enables users to access websites through its complete directory system and hierarchical domain name lookup process. My understanding of global communication systems improved after I learned to track the complete path of requests from local caches to different name servers until they reached their final destination.

Link to Module Completion

https://tryhackme.com/p/ngugiallan05